Who structures the offshore network of the Panama Papers?

Graph Analytics 2026 (MSA-DAT07-05), project report

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Code and data instructions: https://github.com/achillekrtf/offshore-graph

1. Introduction and question

The Panama Papers leak (2016) exposed the client files of the law firm Mossack Fonseca: thousands of offshore companies, the people behind them and the intermediaries that ordered them. The International Consortium of Investigative Journalists (ICIJ) published the structure as a graph.

Question. Who structures the offshore network of the Panama Papers? We identify the central intermediaries, officers and jurisdictions (degree, betweenness, PageRank), then check whether the communities found by Louvain follow the jurisdictions of the entities, the countries of the officers, or the intermediaries.

Why this dataset. The ICIJ database is a public, documented graph of real offshore structures. It is large enough to need modeling decisions (hubs, bipartite structure, isolated nodes) and it carries attributes (jurisdiction, country, leak of origin) against which a structural result can be checked. The question is a structural one: which actors hold the network together, and does its community structure reflect geography or the intermediaries that built it.

Code repository. https://github.com/achillekrtf/offshore-graph (analysis code in `src/` and `notebooks/`, instructions in the README).

The group has three members, with the approval of the course, which allowed an exception to the usual limit of two.

2. Data

Source. ICIJ Offshore Leaks Database, bulk CSV (`full-oldb.LATEST.zip`, generated 2026-09-09), https://offshoreleaks.icij.org/pages/database, downloaded 2026-11-06. License: the database is under the Open Database License (ODbL 1.0) and its contents under Creative Commons Attribution-ShareAlike 3.0. ICIJ asks users to cite the International Consortium of Investigative Journalists.

Structure. The database is a graph stored in CSV files, one per node type plus one for links, joined on a unique `node_id`. The column `sourceID` gives the leak of origin and lets us keep the Panama Papers.

Node type	Meaning	Count (Panama Papers)
Entity	Offshore company, trust or foundation	213,634
Officer	Person or company that owns, directs or fronts an entity (including nominees)	238,402
Intermediary	Law firm, bank or agent that ordered the structure	14,110
Address	Postal address	93,454

Three relationship types connect an entity to the rest of the network: `officer_of` (309,363 links), `intermediary_of` (213,634) and `registered_address` (151,105). The other types in the database (`same_name_as`, `similar`...) are deduplication hints from ICIJ and are ignored. Most `registered_address` links go from an officer to an address (145,795 of 151,105), so addresses add little to the entity graph.

3. Graph model

Bipartite graph. Entities on one side, connectors (officers, intermediaries, entity addresses) on the other. A link says "this connector is attached to this entity". The graph is undirected, simple and unweighted. We use a simple graph and not a multigraph because a connector is linked to an entity at most once after removing duplicate rows, and the projection keeps the multiplicity of shared connectors as an edge weight. The graph is undirected: only the existence of a link matters for our questions, and the direction of the ICIJ links (connector to entity) carries no further information. It has 468,623 nodes and 511,687 edges.

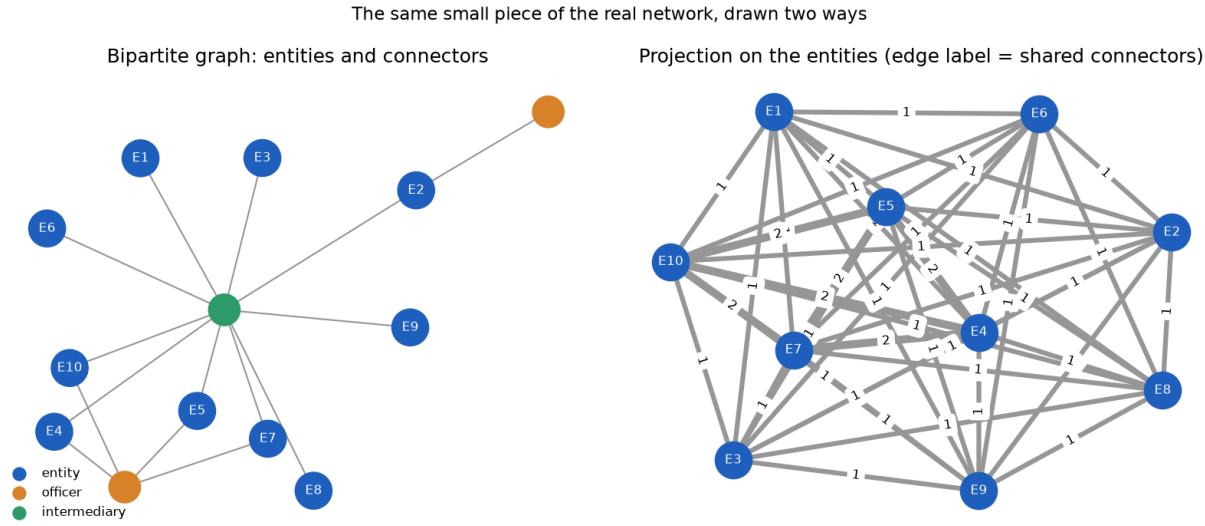
Entity projection. Two entities are linked when they share a connector. The weight is the number of shared connectors. A connector attached to *d* entities adds *d(d-1)/2* edges, so the intermediary with 7,016 entities alone would add about 24.6 million edges. We cap the hubs in the projection: officers with more than 500 entities and intermediaries or addresses with more than 50 are ignored when linking entities. This gives 213,634 nodes and 1,238,967 weighted edges.

Sensitivity of the intermediary cap. We rebuilt the projection with intermediary caps of 10, 50 (our choice), 100 and 500.

Intermediary cap	Edges	Isolated entities	Largest component
10	857,671	153,511 (71.9 %)	18,901 (8.8 %)
50	**1,238,967**	**129,607 (60.7 %)**	**26,643 (12.5 %)**
100	2,060,376	111,417 (52.2 %)	34,705 (16.2 %)
500	9,524,444	66,675 (31.2 %)	73,218 (34.3 %)

A higher cap links more entities and grows the giant component, at the price of many more edges (7.7 times more at 500 than at 50) and of cliques created by a few big firms. The share of isolated entities

stays between 31 % and 72 %, so the fragmentation of the projection is a property of the data and the cap only moves it. We keep 50 as a compromise between coverage and artificial cliques.



One small piece of the real network: bipartite graph (left) and its projection on the entities (right)

The figure shows a real piece of the network with 10 entities, 2 officers and 1 intermediary. In the bipartite graph (13 nodes, 15 edges), the intermediary is linked to all 10 entities and one officer to four of them. In the projection (10 nodes, 45 edges), every pair of entities is linked because they share the intermediary, and the 6 pairs among the four entities that also share the officer have weight 2. One intermediary linked to 10 entities creates 10 x 9 / 2 = 45 edges, which is why the hubs are capped.

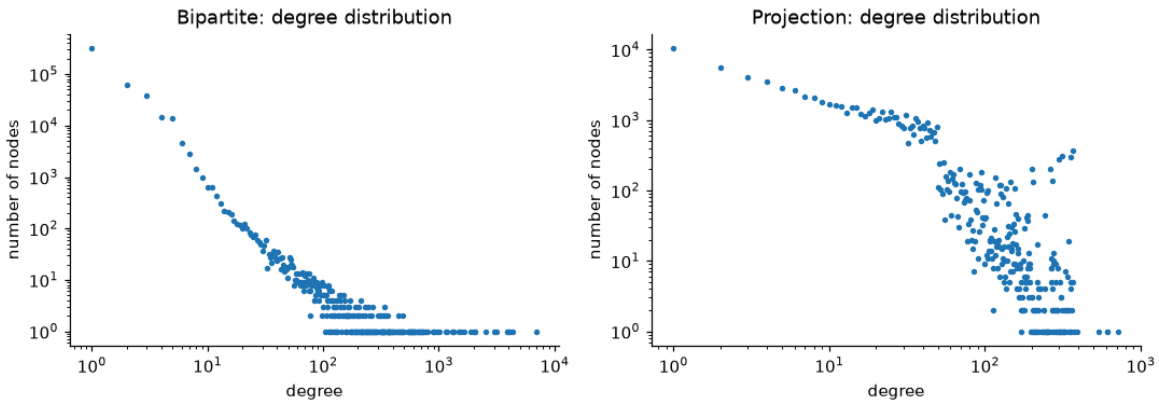
Choices that matter.

- Centralities are computed on the bipartite graph, where intermediaries are nodes, and on the projection, where only entities are.
- Communities are computed on the projection.
- The cap removes the largest hubs from the projection. This lengthens distances and isolates entities that only share a hub connector.

4. Basic statistics and connectivity

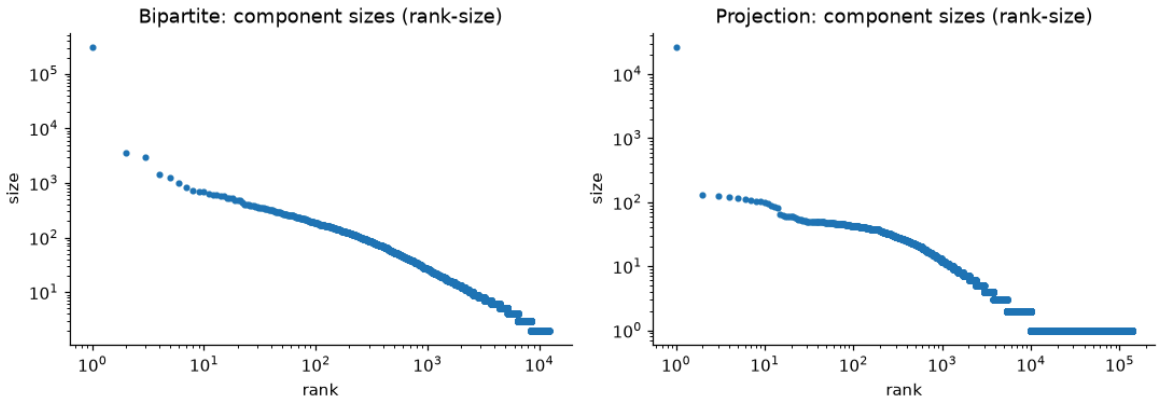
	Bipartite	Entity projection
Nodes	468,623	213,634
Edges	511,687	1,238,967
Density	4.7e-6	5.4e-5
Mean degree	2.18	11.6
Max degree	7,016	720
Connected components	12,218	139,643
Largest component	310,904 (66.3 %)	26,643 (12.5 %)
Isolated nodes	0	129,607

Neither graph is connected. The bipartite graph has 12,218 components and the projection 139,643. Distances are therefore defined per component (section 5), and betweenness and community detection are run on the largest component or on the non-isolated part, as stated in each section.



Degree distributions, log-log

Degrees follow a heavy-tailed distribution: most nodes have one or two links and a few hubs have thousands. In the projection, 129,607 entities (61 %) are isolated: their connectors are shared by nobody, or exceed the cap. The bipartite graph has one giant component and 12,217 small ones.



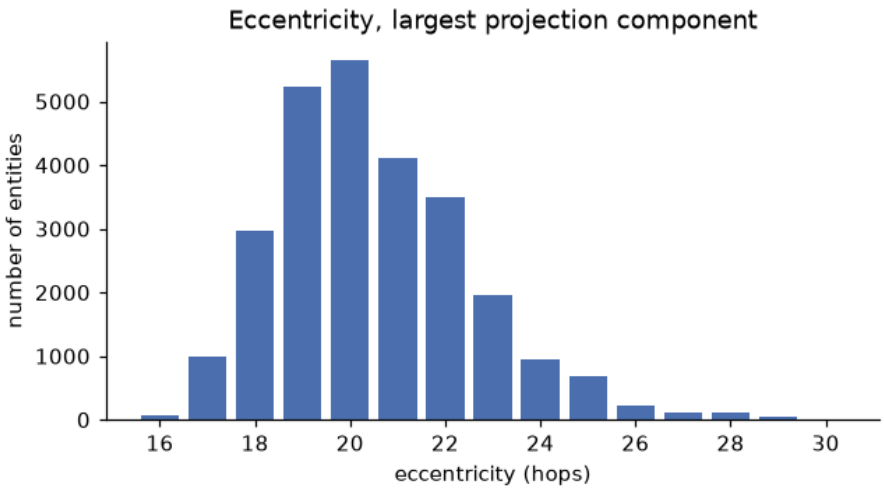
Component sizes (rank-size)

5. Diameter and radius

Distances are in hops, per connected component, since both graphs are disconnected. The diameter is the largest eccentricity and the radius the smallest.

- **Projection:** exact, with a breadth-first search from every node (scipy `csgraph`, 109 s for the largest component).
- **Bipartite giant component** (310,904 nodes): all-pairs is too heavy, so we report a lower bound on the diameter (20 double sweeps) and an upper bound on the radius (eccentricity of the midpoint of the longest path found).

Graph	Component	Nodes	Diameter	Radius
Projection	1st	26,643	30	16
Projection	2nd to 5th	115 to 132	5 to 9	3 to 5
Bipartite	Giant	310,904	at least 37	at most 20

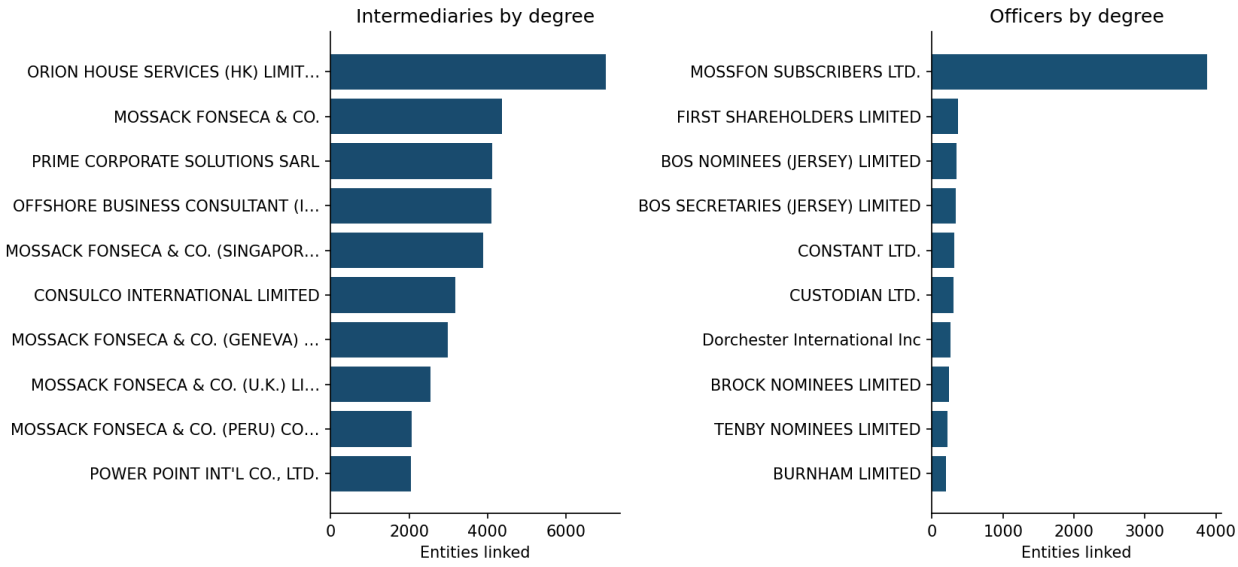


Eccentricity in the largest projection component

In the largest projection component, eccentricities are centered around 20 and reach 30, with 74 central nodes of eccentricity 16. A path of length 2 in the bipartite graph (entity, connector, entity) equals one hop in the projection, so the two sets of distances are not comparable directly.

6. Centralities

We compute degree centrality, PageRank (damping 0.85) and betweenness on the bipartite graph and on the projection. Betweenness is estimated by sampling sources (50 to 300 on the 311,000-node bipartite giant component, 200 on the projection component), because exact betweenness needs a shortest-path tree from every node.



Highest-degree intermediaries and officers

Intermediaries. The table gives the ten intermediaries with the most entities, with the three measures. Degree is the number of entities a firm is attached to. PageRank is shown as a multiple of the average

node score. Betweenness is the sampled share of shortest paths through the firm, from 300 sources, with its rank among the 1,762 intermediaries of the giant component.

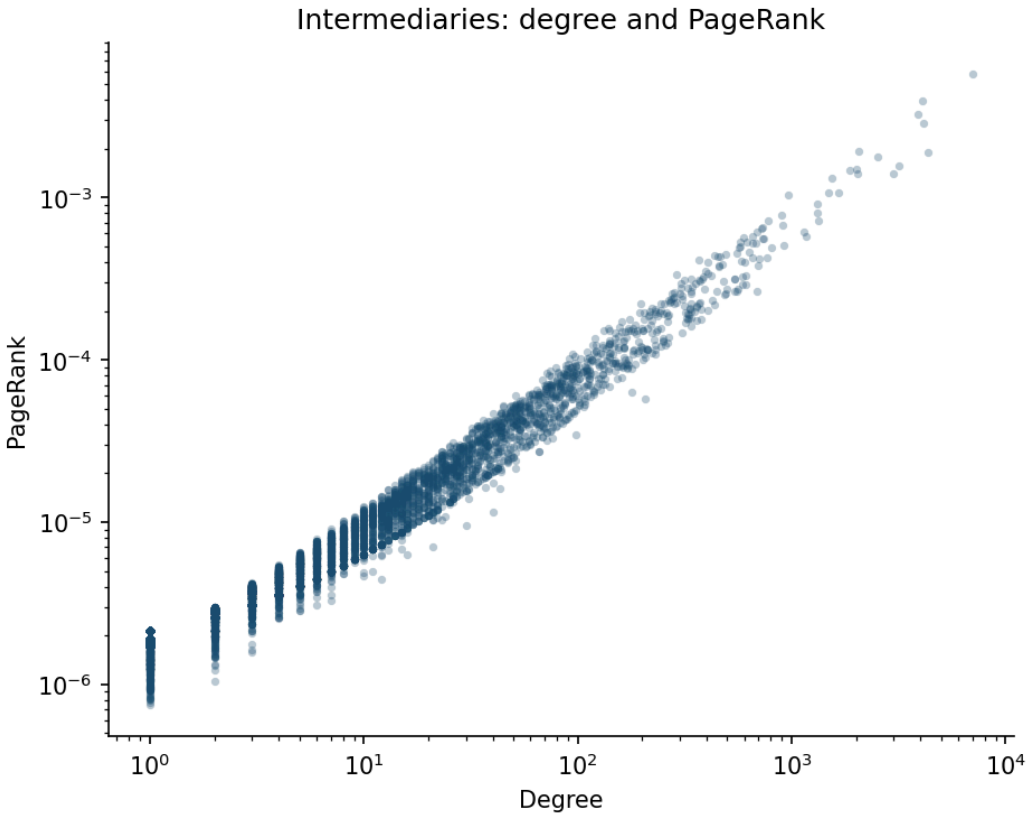
Intermediary	Country	Entities	PageRank (x average)	Betweenness	Rank
Orion House Services (HK) Limited	Hong Kong	7,016	2,703	0.101	4
Mossack Fonseca & Co.	Panama	4,364	887	0.110	3
Prime Corporate Solutions SARL	Luxembourg	4,117	1,343	0.067	7
Offshore Business Consultant (Int'l) Limited	Hong Kong	4,094	1,839	0.097	5
Mossack Fonseca & Co. (Singapore) Pte Ltd.	Singapore	3,888	1,535	0.144	1
Consulco International Limited	UAE, Cyprus	3,168	732	0.043	11
Mossack Fonseca & Co. (Geneva) S.A.	Switzerland	2,986	659	n/a	n/a
Mossack Fonseca & Co. (U.K.) Limited	United Kingdom	2,538	833	0.138	2
Mossack Fonseca & Co. (Peru) Corp.	Peru	2,055	900	0.030	13
Power Point Int'l Co., Ltd.	Taiwan	2,045	658	0.012	40

The top 15 by degree and by PageRank are the same 15 names, and the rank correlation between the two scores over all intermediaries is 0.92.

What the three measures say. Degree counts a firm's clients, and Orion House has the most. PageRank also rewards links to well-connected nodes, and Orion House scores 2,703 times the average node. Betweenness measures brokerage, and the measures disagree in a useful way. Orion House is only 4th by betweenness. Horizon Investment SA (Switzerland) has 540 entities (57th by degree) but ranks 6th by betweenness: it sits on many shortest paths despite a small client base. At the other end, the Peru office (2,055 entities) ranks 13th and Power Point Int'l (Taiwan, 2,045 entities) 40th: volume without brokerage. The Geneva office has no estimate because it lies outside the giant component.

Stability of the sampled betweenness. We repeated the estimate with five random samples of 50 sources and once with 300. The Singapore and UK offices of Mossack Fonseca are in the top 3 in all six runs. The other places move: Orion House ranks between 2nd and 12th in the 50-source runs and 4th with 300 sources, and the top 6 of a 50-source run shares between 4 and 6 names with the 300-source top 6. We therefore read the betweenness ranking as a set (the Singapore, UK and Panama offices of

Mossack Fonseca, Orion House, Offshore Business Consultant, Horizon Investment) and report the 300-source ranks.

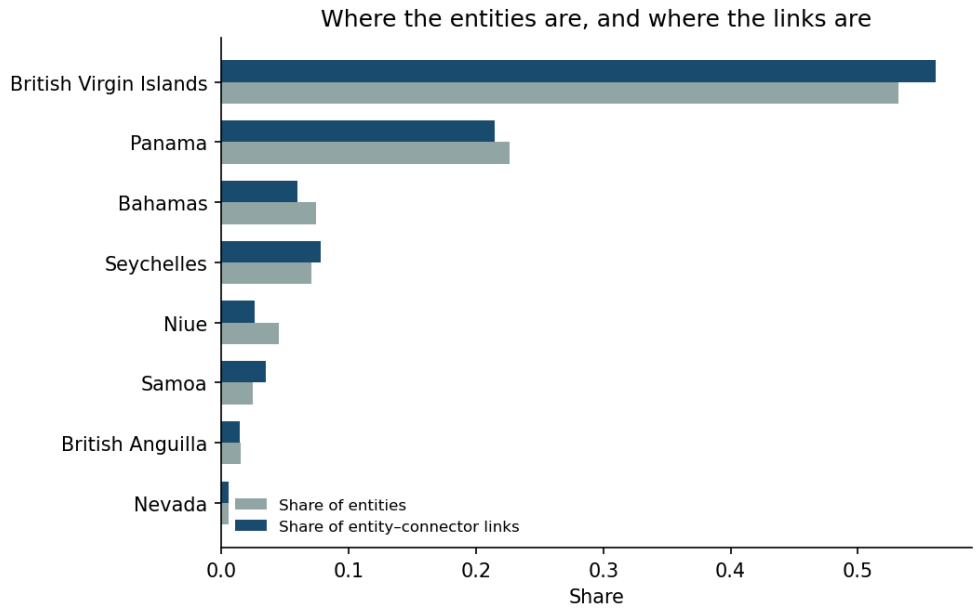


Intermediary degree against PageRank

Officers. Mossfon Subscribers Ltd. (Samoa) is attached to 3,882 entities, against 368 for the next officer. It is a nominee company, and in a bipartite graph every path between two entities goes through a connector, so this single node carries a large share of the shortest paths.

Projection. On the projection, rank correlations between degree, PageRank and betweenness are weak (Spearman 0.2 to 0.65), and none of the 20 most central entities of the bipartite graph appears in the top 20 of the projection. The two graphs answer different questions: the bipartite graph ranks the actors that hold the links, the projection ranks entities that sit between many others.

Jurisdictions. Jurisdictions are node attributes, not nodes. The British Virgin Islands hold 113,648 entities, Panama 48,360, the Bahamas 15,915 and the Seychelles 15,182. Among the 50 entities with the highest bipartite PageRank, 37 are in the British Virgin Islands.



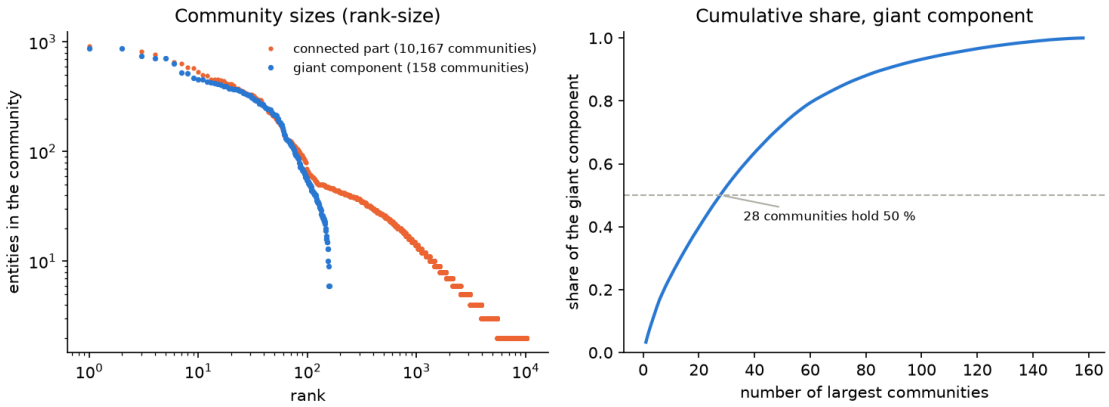
Jurisdiction shares

7. Communities

Method. Louvain (Blondel et al., 2008) maximizes the weighted modularity of the entity projection, with resolution 1.0. The 129,607 isolated entities cannot belong to any community and are excluded, which leaves 84,027 entities: one giant component of 26,643 entities and 10,035 small components of 2 to 132 entities. The giant component is the main result. A small component of two entities sharing one officer is trivially one pure community and would inflate every score.

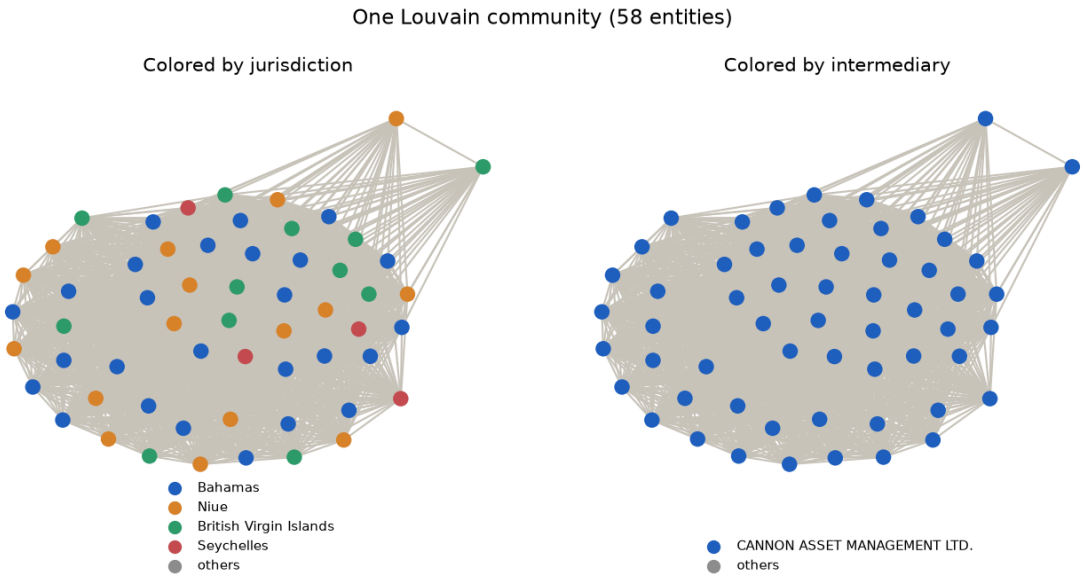
Stability. Five seeds give 158 to 160 communities on the giant component, a modularity of 0.957 and a mean pairwise NMI of 0.98 between seeds. python-louvain gives 158 communities, modularity 0.957 and an NMI of 0.99 with the NetworkX partition.

Shape. The giant component splits into 158 communities, the largest with 875 entities and a median of 92. Twenty-eight communities hold half of the entities and 75 communities of at least 100 entities hold 86 %. A modularity of 0.957 describes dense blocks joined by few edges, which matches the diameter of 30. On the whole connected part, 10,034 of the 10,035 small components are exactly one community each.



Community sizes

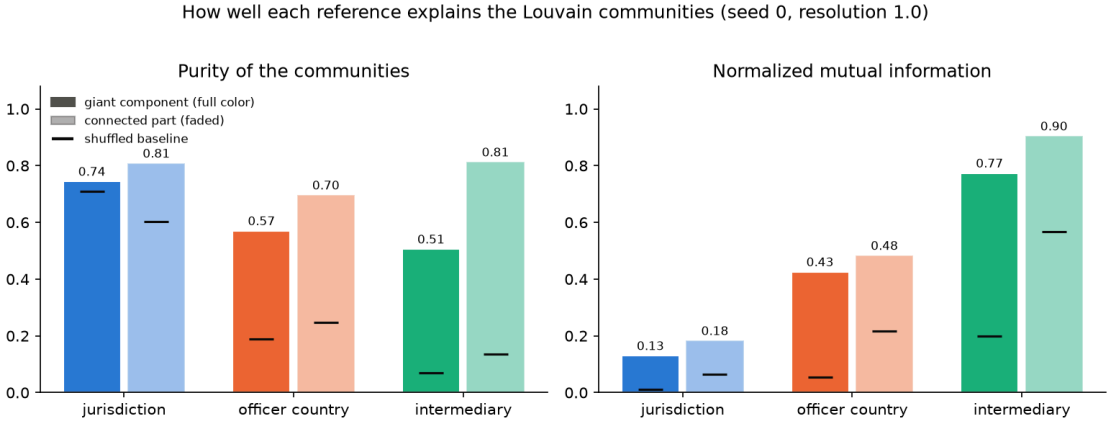
A real community. The figure shows one community of the giant component, drawn twice with the same layout. It has 58 entities served by a single intermediary, Cannon Asset Management Ltd., and registered in four jurisdictions (Bahamas 26, Niue 16, British Virgin Islands 12, Seychelles 4). Colored by jurisdiction, the community looks mixed. Colored by intermediary, it is uniform. Its entities are tied together through officers they share, and those officers belong to the same client book. This is the pattern behind the numbers below.



One Louvain community, colored by jurisdiction and by intermediary

Which reference explains the communities. Each entity carries three labels: its jurisdiction, the most frequent country of its officers and its intermediary. We measure agreement with purity, inverse purity and normalized mutual information (NMI), against a baseline where the labels are shuffled over the entities. The baseline matters because the British Virgin Islands hold 71 % of the giant component, so a random partition already looks pure for the jurisdiction.

Reference (giant component)	NMI	NMI of shuffled labels	Purity	Inverse purity
Intermediary	0.77	0.20	0.51	0.79
Country of the officers	0.43	0.06	0.57	0.21
Jurisdiction	0.13	0.01	0.74 (baseline 0.71)	0.07



Agreement of the communities with each reference

NMI ranks the intermediary first, the country of the officers second and the jurisdiction last. Jurisdiction purity sits only 0.04 above its shuffled baseline. Inverse purity shows that 79 % of the entities sit in the community where most of their intermediary's clients sit, against 21 % for the officer country and 7 % for the jurisdiction. The communities therefore follow the client books of the intermediaries and, to a lesser extent, groups of officers from the same country. The order holds at every resolution from 0.25 to 4. A community is usually larger than one intermediary: the dominant intermediary covers 51 % of its community on average, and the largest communities group several firms from the same place, such as Jersey trust companies or Panama law firms.

Effect of the hub cap. 65 % of the giant component's entities (17,278) have an intermediary above the cap of 50, so their intermediary created no edge. They still cluster by intermediary (NMI 0.80, purity 0.70), through the nominee officers and addresses that the clients of a big firm share. The cap removes the mechanical effect of a shared intermediary and the result remains.

Conclusion. The hypothesis holds: the communities of the Panama Papers network follow the intermediaries more than the officers' countries and far more than the jurisdictions. The match is partial: the dominant intermediary covers 51 % of its community on average, so a community often groups several firms from the same place, and a single firm can be spread over several communities.

8. Discussion and limits

- One leak only: no comparison across leaks, and the structure reflects one law firm's clients.
- Hub cap: a modeling choice that changes distances and isolates 61 % of entities in the projection (31 % to 72 % depending on the cap, section 3).
- Sampled betweenness: the order changes with the sample, only the Singapore and UK offices of Mossack Fonseca stay in the top 3 in every run (section 6).
- Communities are computed on the 39 % of entities that are not isolated in the projection. The 61 % left out have no information to cluster on.
- The bulk file reflects ICIJ's own deduplication and a data cut-off in 2015.
- Names refer to companies and persons published by ICIJ. Presence in the leak does not imply wrongdoing.

9. Reproducibility, code and AI tools

- **Software and methods:** all analyses are run in NetworkX 3.3 (Python 3.12) with pandas, scipy and python-louvain. Louvain uses the NetworkX implementation, cross-checked once with python-louvain. Neo4j (local instance) was used only for a dashboard of the centrality rankings and to check the intermediary degrees with Cypher queries (`neo4j/`). Every parameter (hub caps, sampling sizes, resolution, seeds) is stated in the sections above and fixed in the notebooks, so the analysis can be reproduced.
- **Code:** Repository: https://github.com/achillekrtf/offshore-graph (`src/` for the code, `notebooks/` for the analysis, README for the steps to reproduce). The raw data is not stored in the repository.
- **Reused code:** no existing code for this dataset was reused. Apart from the libraries above and the algorithms cited in the references, all code was written by the group.
- **AI tools used and purpose:** the three authors used AI assistants, all for coding help and drafting, never as a source of results.
- Claude Code (Anthropic), used by all three: Achille Krotoff for the loading and graph-construction code, the diameter computation, the cap sensitivity test, notebooks 01 and 02 and a first draft of this report; Jad Yousri and Adrian Ravas for coding help and debugging in notebooks 03 and 04.

- Cursor, used by Jad Yousri and Adrian Ravas for coding help and debugging in notebooks 03 and 04, and for the Neo4j import and dashboard of notebook 03.
- Every number in the report comes from code that was run, and the authors checked the outputs and the interpretations.

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